

Photoelectric Effect & Photon Energy — Lecture Note

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Topic: Photon energy, threshold frequency, work function, stopping potential, intensity vs photon count. Problems 7–15 from class handout.

⚡ How to use this note

Start by reading the **Notes** (key formulas + constants), then work through the two **Examples**. After that, attempt each Problem 7–15 in the dashed work-space first — *don't peek at the green Solution box until you've tried*. Final answers are boxed in green.

■ Notes — Photoelectric Effect Toolkit

1. Photon energy

A single photon of light of frequency f (wavelength λ) carries energy

$$E = hf = \frac{hc}{\lambda}.$$

2. Wave relation

$$c = f\lambda \iff f = \frac{c}{\lambda}, \lambda = \frac{c}{f}.$$

3. Work function W_0

Minimum energy needed to free an electron from a metal surface (a property of the metal).

The **threshold** frequency and wavelength satisfy

$$hf_0 = W_0, \quad \lambda_0 = \frac{hc}{W_0}.$$

4. Einstein's photoelectric equation

For a photon with energy *greater* than W_0 , the ejected electron carries the surplus as kinetic energy:

$$KE_{\max} = hf - W_0.$$

5. Stopping potential

The voltage V_s that just stops the fastest electrons:

$$eV_s = KE_{\max} = hf - W_0.$$

So V_s (in volts) numerically equals $KE_{\max}$ (in eV).

6. Intensity vs photon count

Power P delivered by a beam = (energy per photon) $\times$ (photons per second):

$$N = \frac{P}{E_{\text{photon}}}, \quad \text{Intensity } I = \frac{P}{A}.$$

7. Diffraction grating

For a grating with d = distance between adjacent slits,

$$d \sin \theta_m = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots).$$

Distance on a screen at L : $y_m = L \tan \theta_m$.

Useful constants and conversions:

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s} \cdot c = 3.00 \times 10^8 \text{ m/s} \cdot e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J} \cdot hc \approx 1.988 \times 10^{-25} \text{ J}\cdot\text{m} \approx 1240 \text{ eV}\cdot\text{nm}$$

Worked Examples

Example 1 — Photon energy from wavelength

A laser pointer emits red light at $\lambda = 650$ nm. What is the energy of one photon? Give the answer in both joules and electron-volts.

Idea. Use $E = hc/\lambda$. The shortcut $hc \approx 1240 \text{ eV}\cdot\text{nm}$ makes the eV form trivial.

Step 1 — In joules. $E = \frac{(6.626 \times 10^{-34})(3.00 \times 10^8)}{650 \times 10^{-9}} = \frac{1.988 \times 10^{-25}}{6.50 \times 10^{-7}} = 3.06 \times 10^{-19} \text{ J}.$

Step 2 — In eV. $E = \frac{1240 \text{ eV}\cdot\text{nm}}{650 \text{ nm}} = 1.91 \text{ eV}.$

Answer. $E \approx 3.06 \times 10^{-19} \text{ J} \approx 1.91 \text{ eV}$

Example 2 — Threshold + Stopping potential

Sodium has a work function $W_0 = 2.28$ eV. Light of $\lambda = 400$ nm strikes a sodium surface. (a) Find the threshold wavelength. (b) Find the stopping potential V_s .

(a) Threshold wavelength. $\lambda_0 = \frac{hc}{W_0} = \frac{1240 \text{ eV}\cdot\text{nm}}{2.28 \text{ eV}} \approx 544 \text{ nm}.$ Light with $\lambda > 544$ nm cannot eject electrons from Na.

(b) Photon energy at 400 nm. $E = 1240/400 = 3.10 \text{ eV}.$

Stopping potential. $eV_s = E - W_0 = 3.10 - 2.28 = 0.82 \text{ eV} \Rightarrow V_s = 0.82 \text{ V}.$

Answers. $\lambda_0 \approx 544 \text{ nm}, V_s \approx 0.82 \text{ V}$

Problems 7 – 15

Problem 7 Visible-light frequency & photon-energy range

STATEMENT

According to Wikipedia, the human eye is sensitive to light with wavelengths between **380 nm** and **750 nm**.

1. What range of frequencies of radiation does this correspond to?
2. What range of photon energies does this correspond to?

WORK SPACE – YOUR SOLUTION

SOLUTION

(a) Frequencies. Use $f = c/\lambda$.

$$f_{\text{at } 750\text{nm}} = \frac{3.0 \times 10^8}{7.50 \times 10^{-7}} = 4.00 \times 10^{14} \text{ Hz}$$

$$f_{\text{at } 380\text{nm}} = \frac{3.0 \times 10^8}{3.80 \times 10^{-7}} = 7.89 \times 10^{14} \text{ Hz}$$

$$f \in [4.00 \times 10^{14}, 7.89 \times 10^{14}] \text{ Hz}$$

(b) Photon energies. $E = hc/\lambda = 1240 \text{ eV}\cdot\text{nm}/\lambda$:

$$E_{750} = 1240/750 = 1.65 \text{ eV} \approx 2.65 \times 10^{-19} \text{ J}$$

$$E_{380} = 1240/380 = 3.26 \text{ eV} \approx 5.23 \times 10^{-19} \text{ J}$$

$$E \in [1.65, 3.26] \text{ eV} \approx [2.65, 5.23] \times 10^{-19} \text{ J}$$

Mnemonic. Red end = lower frequency, lower energy. Violet end = higher frequency, higher energy.

Problem 8 Work function $W_0 = 4.8 \times 10^{-19} \text{ J}$

STATEMENT

A photoemissive surface has a work function of $4.8 \times 10^{-19} \text{ J}$.

1. What is the minimum photon energy needed to cause electrons to be emitted from the surface?
2. Determine the threshold frequency and wavelength for photoemission from this surface.

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Min photon energy. Equal to the work function: $E_{\min} = W_0$

$$E_{\min} = 4.8 \times 10^{-19} \text{ J} \approx 3.00 \text{ eV}$$

(b) Threshold.

$$f_0 = W_0/h = (4.8 \times 10^{-19})/(6.626 \times 10^{-34}) = 7.24 \times 10^{14} \text{ Hz}$$

$$\lambda_0 = c/f_0 = (3.0 \times 10^8)/(7.24 \times 10^{14}) = 4.14 \times 10^{-7} \text{ m}$$

$$f_0 \approx 7.24 \times 10^{14} \text{ Hz}, \lambda_0 \approx 414 \text{ nm}$$

Sanity check. 414 nm is violet light — makes sense, this metal needs energetic photons.

Problem 9 Work function $W_0 = 3.0 \text{ eV}$ **STATEMENT**

A photoemissive surface has a work function of **3.0 eV**.

1. Determine the threshold frequency and wavelength for this material.
2. Photons cause electrons to be emitted with a stopping voltage of **2.0 V**. Determine the energy of the photons and the wavelength of the light.

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) **Threshold.** $\lambda_0 = hc/W_0 = 1240 \text{ eV} \cdot \text{nm} / 3.0 \text{ eV} = 413 \text{ nm}$.

$f_0 = c/\lambda_0 = 3.0 \times 10^8 / 4.13 \times 10^{-7} = 7.25 \times 10^{14} \text{ Hz}$.

$$f_0 \approx 7.25 \times 10^{14} \text{ Hz}, \lambda_0 \approx 413 \text{ nm}$$

(b) With $V_s = 2.0 \text{ V}$. $KE_{\text{max}} = eV_s = 2.0 \text{ eV}$. So

$E_{\text{photon}} = KE_{\text{max}} + W_0 = 2.0 + 3.0 = 5.0 \text{ eV} = 8.01 \times 10^{-19} \text{ J}$.

$\lambda = 1240/5.0 = 248 \text{ nm}$.

$$E = 5.0 \text{ eV} (8.01 \times 10^{-19} \text{ J}), \lambda \approx 248 \text{ nm}$$

Problem 10 He-Ne laser, $\lambda = 633 \text{ nm}$, $P = 0.50 \text{ W}$ **STATEMENT**

A He-Ne laser has a wavelength of **633 nm** and an output power of **0.50 W**.

1. What is the photon energy for this light?
2. The laser hits a black target. How many photons hit the target per second?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Photon energy. $E = hc/\lambda = 1240/633 = 1.96 \text{ eV} \approx 3.14 \times 10^{-19} \text{ J}$.

$$E \approx 1.96 \text{ eV } (3.14 \times 10^{-19} \text{ J})$$

(b) Photons per second. $N = P/E = 0.50/3.14 \times 10^{-19}$.

$$N \approx 1.59 \times 10^{18} \text{ photons/s}$$

Insight. Even a small 0.5 W laser emits about 10^{18} photons every second — light is "lumpy" at this scale, but the lumps are so numerous it looks continuous.

Problem 11 $\lambda = 440 \text{ nm}$ grating with 10,000 lines/cm**STATEMENT**

Light of wavelength **440 nm** is incident on a grating with **10,000 lines/cm**.

1. What is the angular deflection of the first and second order maxima?
2. What are the positions of these maxima on a screen that is **1.2 m** from the grating?
3. Suppose it is desired to detect the maxima with a photocell. What is the maximum allowable work function of the cell if it is to respond to the light?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Angles. Slit spacing $d = 1/(10,000 \text{ cm}^{-1}) = 1.0 \times 10^{-6} \text{ m}$. Use $d \sin \theta_m = m\lambda$:

$$\sin \theta_1 = 440 \times 10^{-9} / 1.0 \times 10^{-6} = 0.44 \Rightarrow \theta_1 \approx 26.1^\circ.$$

$$\sin \theta_2 = 0.88 \Rightarrow \theta_2 \approx 61.6^\circ.$$

$$\theta_1 \approx 26.1^\circ, \theta_2 \approx 61.6^\circ$$

(b) Positions at $L = 1.2 \text{ m}$. $y_m = L \tan \theta_m$:

$$y_1 = 1.2 \tan(26.1^\circ) = 1.2 \times 0.490 = 0.588 \text{ m}.$$

$$y_2 = 1.2 \tan(61.6^\circ) = 1.2 \times 1.847 = 2.22 \text{ m}.$$

$$y_1 \approx 0.59 \text{ m}, y_2 \approx 2.22 \text{ m}$$

(c) Max work function. For the photocell to fire,

$$W_0 \leq E_{\text{photon}} = hc/\lambda = 1240/440 = 2.82 \text{ eV}.$$

$$W_0 \leq 2.82 \text{ eV } (\approx 4.52 \times 10^{-19} \text{ J})$$

Problem 12 Silver — threshold $\lambda_0 = 262 \text{ nm}$

STATEMENT

The threshold wavelength for the photoelectric effect for silver is **262 nm**.

1. Find the work function for silver.
2. Find the stopping potential if the incident radiation has a wavelength of **200 nm**.

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Work function. $W_0 = hc/\lambda_0 = 1240/262 = 4.73 \text{ eV} \approx 7.59 \times 10^{-19} \text{ J}$.

$$W_0 \approx 4.73 \text{ eV } (7.59 \times 10^{-19} \text{ J})$$

(b) Stopping potential @ $\lambda = 200 \text{ nm}$. $E_{\text{photon}} = 1240/200 = 6.20 \text{ eV}$.

$$eV_s = E - W_0 = 6.20 - 4.73 = 1.47 \text{ eV}.$$

$$V_s \approx 1.47 \text{ V}$$

Problem 13 $\lambda = 400 \text{ nm}$ beam, intensity 100 W/m^2 **STATEMENT**

A light beam of wavelength **400 nm** has an intensity of **100 W/m²**.

1. What is the energy of each photon in the beam?
2. How much energy strikes an area of **1.0 cm²** perpendicular to the beam in **1.0 s**?
3. How many photons strike this area in **1.0 s**?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Photon energy. $E = 1240/400 = 3.10 \text{ eV} \approx 4.97 \times 10^{-19} \text{ J}$.

$$E \approx 3.10 \text{ eV } (4.97 \times 10^{-19} \text{ J})$$

(b) Energy on 1 cm² in 1 s. $1 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$.

$$U = I A t = 100 \times 1 \times 10^{-4} \times 1.0 = 1.00 \times 10^{-2} \text{ J} = 10 \text{ mJ}.$$

$$U = 0.010 \text{ J}$$

(c) Photons/sec. $N = U/E = 1.00 \times 10^{-2} / 4.97 \times 10^{-19}$.

$$N \approx 2.01 \times 10^{16} \text{ photons}$$

Problem 14 **Cesium** — $W = 1.9 \text{ eV}$ **STATEMENT**

The work function for cesium is **1.9 eV**.

1. Find the threshold frequency and wavelength for the photoelectric effect.
2. Find the stopping potential if the wavelength of incident light is **300 nm**.

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Threshold. $\lambda_0 = 1240/1.9 \approx 653 \text{ nm}$.

$$f_0 = c/\lambda_0 = 3.0 \times 10^8 / 6.53 \times 10^{-7} \approx 4.59 \times 10^{14} \text{ Hz}.$$

$$f_0 \approx 4.59 \times 10^{14} \text{ Hz}, \lambda_0 \approx 653 \text{ nm}$$

(b) Stopping potential @ $\lambda = 300 \text{ nm}$. $E = 1240/300 = 4.13 \text{ eV}$.

$$eV_s = E - W = 4.13 - 1.9 = 2.23 \text{ eV}.$$

$$V_s \approx 2.23 \text{ V}$$

Why cesium? Very low work function $\rightarrow$ fires on visible red/orange light. Used in old TV photodetectors and photoelectric cells.

Problem 15 60 W bulb, 2.1% efficient, $\lambda_{\text{avg}} = 550 \text{ nm}$ **STATEMENT**

A **60.0 W** light bulb is **2.1%** efficient (**2.1%** of the electrical energy is converted into visible light) and the average wavelength of the light is **550 nm**.

1. What is the average energy of a visible photon produced by the light bulb?
2. How many visible photons does the light bulb emit per second?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Photon energy. $E = 1240/550 = 2.25 \text{ eV} \approx 3.61 \times 10^{-19} \text{ J}$.

$$E \approx 2.25 \text{ eV } (3.61 \times 10^{-19} \text{ J})$$

(b) Photons/sec. Visible power = $60.0 \times 0.021 = 1.26 \text{ W}$.

$$N = P_{\text{vis}}/E = 1.26/3.61 \times 10^{-19}.$$

$$N \approx 3.49 \times 10^{18} \text{ photons/s}$$

Comparison. The 60 W bulb spits out about twice as many *visible* photons per second as the 0.5 W laser of Problem 10 — but the bulb's energy goes mostly into infrared (heat), which is why "2.1% efficient" hurts so much.